ClaimWise – Repair & Warranty Assistant

A PROJECT REPORT

Submitted by

MONIKA R

(240701332)

in partial fulfilment of the End Semester Examination for the course

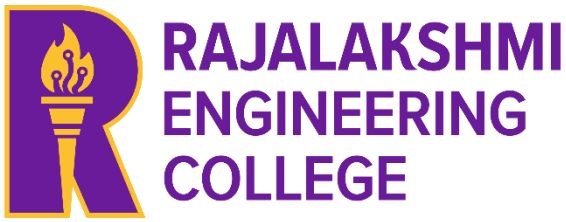
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

In

BACHELOR OF ENGINEERING

[COMPUTER SCIENCE AND ENGINEERING]



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

[NOVEMBER 2026]

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "ClaimWise – Repair & Warranty Assistant" is the bonafide work of MONIKA R (240701332), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester V, under my supervision.

SIGNATURE

Dr.J.MANORANJANI
HEAD OF THE DEPARTMENT
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Dr.J.JINU SOPHIA
COURSE FACULTY
Assistant Professor (SG)
Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on 10-11-2026

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I.NIVETHA S (240701374), hereby declare that the project report titled **"ClaimWise – Repair & Warranty Assistant"** submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 10 November 2026

SIGNATURE OF THE STUDENT
MONIKA R
[240701332]

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MONIKA R
240701332

ABSTRACT

ClaimWise – Repair & Warranty Assistant is a low-code application developed using **Zoho Creator** to simplify and streamline the management of product warranties, warranty claims, and repair requests. Managing product warranty details and repair activities manually can be time-consuming and may lead to difficulties in maintaining records, tracking claim status, and monitoring warranty expiration dates. To address these challenges, ClaimWise provides a centralized platform that organizes product registration, warranty information, claim processing, and repair management in a systematic manner.

The application consists of several modules, including Product Registration, Warranty Claim, Repair Request, and User Profile. The Product Registration module enables users to maintain product details and associated warranty information. The Warranty Claim module allows users to submit claims by providing issue descriptions and tracking their claim status, such as Submitted, Under Review, Approved, Rejected, and Completed. The Repair Request module helps manage repair-related information and monitor repair activities. These modules are connected through lookup fields to maintain relationships between records and improve data organization.

The system supports automatic warranty end-date calculation, unique Claim ID generation, Repair ID generation, and repair cost difference calculation. A warranty reminder workflow is configured to help users keep track of upcoming warranty expirations.

The project is implemented using the low-code development approach offered by Zoho Creator, which provides built-in tools for designing forms, creating reports, configuring workflows, and developing dashboards with minimal traditional programming. Deluge scripting is used wherever additional logic and automation are required.

The primary objective of ClaimWise is to minimize manual effort, improve record management, support timely warranty tracking, and make the repair and claim management process more organized and efficient.

Keywords: Zoho Creator, Low-Code Application Development, Warranty Management, Product Registration, Warranty Claim Processing, Repair Management, Deluge Scripting, Workflow Automation, Reports and Dashboards.

PROJECT AT A GLANCE

Attribute	Details
Project Title	ClaimWise – Repair & Warranty Assistant
Application Domain	Warranty / Product Registration / Repair Management
Student Name & Register No.	Monika R – [240701332]
Zoho Creator Edition	Zoho Creator Trial Edition
Application Name (Link Name)	ClaimWise – Repair & Warranty Assistant
Published / Live URL	https://creatorapp.zoho.in/monika.r.2024.cse_rajalak34/claimwise-repair-management#Section_1
Demo Credentials (per role)	Administrator
Number of Forms / Subforms	5 forms: User Profile, Product Registration, Warranty Claim, Repair Request, Service Provider.
Number of Reports (by type)	9 reports: 5 list view and 4 Kanban reports
Number of Pages / Dashboards	1 Dashboard with key record-count indicators
Number of Workflows	5 workflows
Deluge Functions / Custom Actions	Deluge logic for date, identifier, repair-cost calculations and 1 CRM integration
Integrations	Zoho CRM – Integration workflow configured.
Roles & Profiles Defined	Administrator; registered user/customer
Environments Used	Development / Testing in Zoho Creator
Date of Production Deployment	Not deployed / to be entered if published

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1

INTRODUCTION

1.1 Background

ClaimWise – Repair & Warranty Assistant is developed using Zoho Creator to manage product registration, warranty claims and repair activities in one application. It provides a centralized platform for registered users and administrators to maintain product details, submit warranty claims and monitor repair records.

1.2 Problem Statement

The existing warranty and repair management process can involve manual data collection and tracking, making it difficult to manage submissions and monitor their progress. ClaimWise addresses these challenges through a centralized application management system.

- Difficulty maintaining product and warranty records. • Lack of clear claim and repair status tracking.
- Time-consuming manual reporting and monitoring**.**

1.3 Objectives

- To create product registration forms for submitting registered products and warranty claims.
- To automate application status updates and email notifications.
- To provide reports and dashboards for tracking submissions.
- To connect the application with Zoho Creator for record accessibility.

1.4 Scope of the Project

In Scope:

1. Registration and product registration and claim submission for students, faculty, and external customers.
2. Application tracking through reports, status-based reports, and dashboards.
3. Email notifications and Zoho Creator application.

Out of Scope:

1. Automatic AI-based approval or rejection of products and claims.
2. Full mobile application and customer portal development.
3. Actual funding disbursement and incubation management.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because its drag-and-drop tools help develop ClaimWise quickly with minimal coding. It allows citizen developers to create and maintain forms, reports, and workflows easily. Built-in cloud hosting reduces the need for separate server setup and maintenance. It supports web and mobile access, along with security features and user access controls. Its integrated tools reduce development effort and make the application cost-effective.

1.6 Organisation of the Report

Chapter 1 introduces the project. Chapter 2 reviews existing systems and related literature. Chapter 3 presents the requirements. Chapter 4 explains the system design. Chapter 5 describes implementation. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security, and governance. Chapter 8 presents the results and discussion. Chapter 9 concludes the report and suggests future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing process, students and customers submit their products and claims through manual forms or separate communication channels. The administrator collects the details and maintains the records. Applications are reviewed manually, and customers are informed about their status. Tracking submissions and preparing reports can be time consuming.

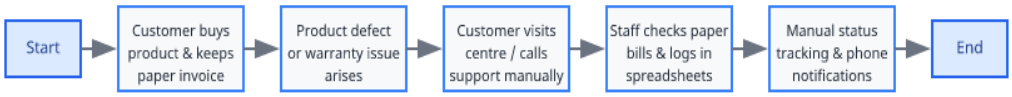


Figure 2.1: [As-Is Process Flow of the Existing System]

2.2 Limitations of the Existing System

Manual data entry makes record maintenance difficult.

Tracking application status and review progress is time-consuming.

Preparing reports and monitoring submissions requires manual effort.

2.3 Low-Code versus Traditional Development

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Takes more Time	Faster Development
Skills required	Programming and coding skills	Basic coding and low-code skills
Hosting & infrastructure	Requires separate hosting setup	Built-in cloud hosting
Mobile application	Requires additional development	Supports mobile access
Security & access control	Must be configured manually	Built-in security and permissions

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Maintenance & change	Requires developer support	Easier to maintain and modify
Customisation limits	Highly flexible	Limited by platform features
Cost	[Generally higher development cost	Generally lower development cost

2.4 Literature Survey

ClaimWise is a Zoho Creator-based warranty and repair management system that helps users register products, submit warranty claims, track claim statuses and manage repair activities.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bock & Frank (2021)	Low-Code Platform	[Low-code improves development productivity and user participation.]	Supports using Zoho Creator
2	Bucaioni et al. (2022)	Modelling in Low-Code Development	Low-code supports visual modelling and application development.	Supports form and application design.
3	Endres et al. (2022)	Warranty and Repair Management Innovation Management	Centralized warranty and repair records support service tracking.	Supports structured product, claim and repair management.
4	[Zoho Corporation (n.d.)	Zoho Creator Documentation	Creator supports forms, reports, and automated workflows.	Supports product registration and status tracking.
5	Jakobsen et al. (2024	Warranty Management Systems	Warranty systems organize product coverage, claims and service records.	Supports ClaimWise workflow and record design.

2.5 Emerging Trends in Low-Code Development

Low-code development is evolving through AI-assisted app building, citizen development, hyper-automation, and composable applications. These trends influenced ClaimWise by encouraging faster form creation, easier application maintenance, and automated status updates and email notifications. Zoho Creator enables nonprogrammers to build applications using visual tools. Future AI integration can help analyse and evaluate submitted warranty claims.

2.6 Proposed System

The proposed ClaimWise system is developed using Zoho Creator to manage product registrations, warranty claims and repair records in one platform. It reduces manual data entry through structured forms, simplifies tracking through status-based reports, and provides dashboard indicators for administrative monitoring.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Manages the ClaimWise application.	Manage product, claim and repair records; review claims; update statuses; monitor reports and dashboard.
Customer / Registered User	Registers products and submits warranty claims.	Maintain profile and product details; submit claims and track claim status.
Service Provider	Provides repair services when assigned.	View assigned repair information and update repair progress, if access is configured.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to maintain profile and product registration details.	Customer	High
FR-02	The system shall calculate or store the warranty end date for a registered product.	System	High
FR-03	The system shall generate a unique Claim ID when a warranty claim is created.	System	High
FR-04	The system shall allow administrators to review claims and update claim status.	Administrator	High
FR-05	The system shall manage repair requests, service providers and assigned repairs.	Administrator	High
FR-06	The system shall calculate repair cost difference and display dashboard counts.	System / Administrato	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports should load within 5 seconds under normal usage.
NFR-02	Security	Users should access only the data permitted by their roles
NFR-03	Usability	The application should be easy to use through a web browser.
NFR-04	Availability	The application should be accessible through Zoho Creator cloud services.
NFR-05	Maintainability	Forms, reports, and workflows should be easy to update
NFR-06	Auditability	Application changes should be traceable wherever audit logging is enabled

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Trial Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho Creator / Zoho Creator /
Mobile	Zoho Creator mobile app (Android / iOS) [if used]
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

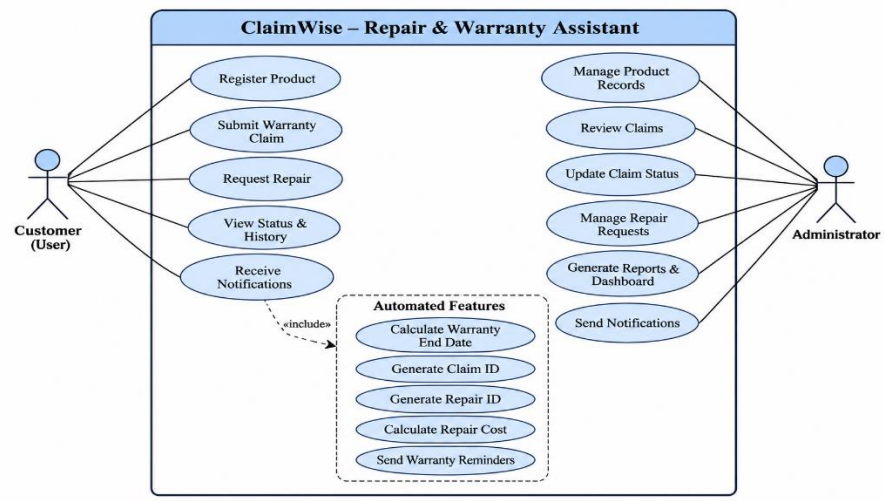


Table 3.5: Use Case Description – Submit Warranty Claim

Use Case ID	UC-01
Use Case Name	Submit Warranty Claim
Actor(s)	Customer / Registered User
Pre-condition	A product is registered and linked to the user profile.
Main Flow	1. User opens Warranty Claim form. 2. Selects the linked product. 3. Enters issue description and required details. 4. Submits the claim. 5. System generates Claim ID and saves the record.
Alternate Flow	If required fields are missing or no product is selected, the user completes the mandatory details before submission.
Post-condition	Warranty claim is saved with a generated Claim ID and an initial claim status.
Related FR	FR-03, FR-04

CHAPTER 4

SYSTEM DESIGN

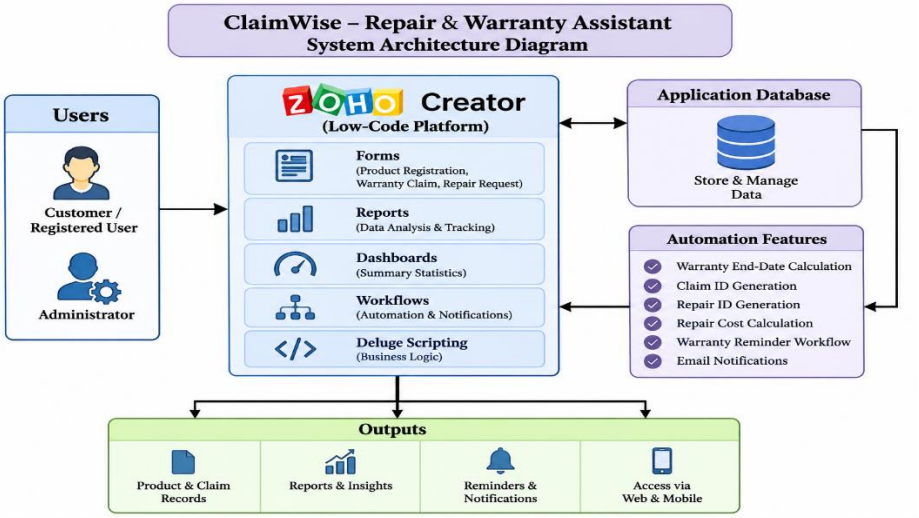


Figure 4.1: System Architecture of the Application

The diagram shows the user layer, Zoho Creator application layer, data layer, and integration layer

4.2 Data Model

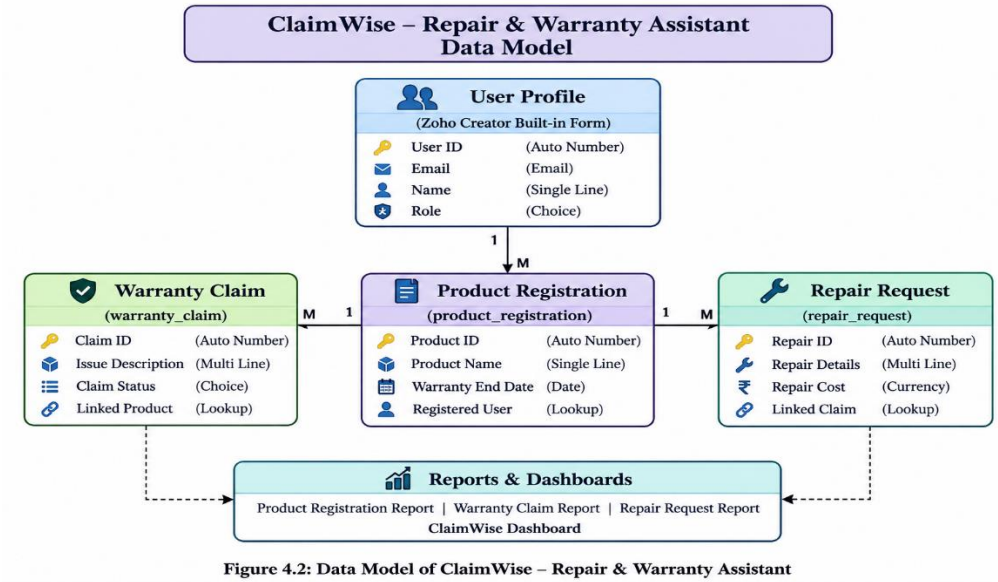


Figure 4.2: Data Model of ClaimWise – Repair & Warranty Assistant

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
User Profile	Name, email and user details	Product Registration / Warranty Claim	One-to-Many	Lookup relationship
Product Registration	product_name, warranty_end_date, registered_user	User Profile	Many-to-One	Lookup field registered_user
Warranty Claim	claim_id, issue_description, claim_status, linked_product	Product Registration	Many-to-One	Lookup field linked_product

4.3 Form Design

Table 4.2: Field Specification – Product Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Product Name	product_name	Single Line	Yes	Required	Registered product details
Warranty End Date	warranty_end_date	Date	As configured	Calculated / entered date	Warranty coverage end date
Registered User	registered_user	Lookup	Yes	Lookup to User Profile	Product owner
Claim ID	claim_id	Single Line	Autogenerated	Unique identifier	Warranty Claim record
Linked Product	linked_product	Lookup	Yes	Lookup to Product Registration	Product associated with claim

4.4 Process and Workflow Design

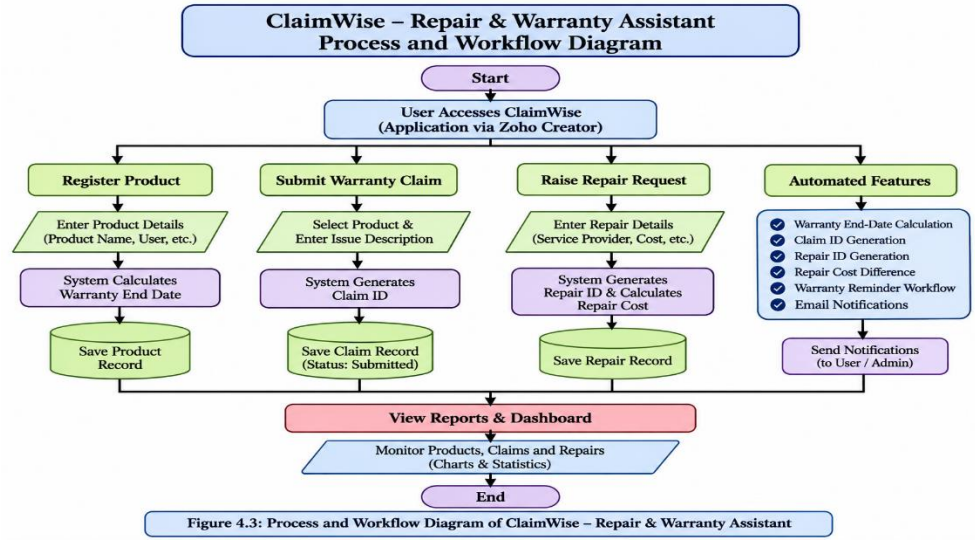


Figure 4.3: Process and Workflow Diagram of ClaimWise – Repair & Warranty Assistant

4.5 Navigation Structure and Wireframe

The diagram below shows the main navigation options and a dashboard layout for the ClaimWise application.

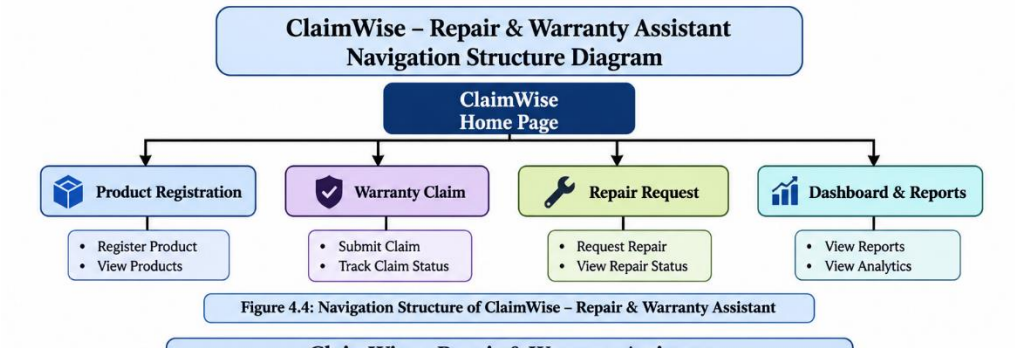


Figure 4.4: Navigation Structure of ClaimWise – Repair & Warranty Assistant



Figure 4.5: Wireframe of ClaimWise Application Pages

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Customer	Service Provider
Product Registration Form	A E V D	A V	-
Warranty Claim Form / Report	A E V D	A V	-
Repair Request / Assigned Repair	A E V D	-	V (if permitted)
Dashboard	V	-	-
Service Provider Records	A E V D	-	V (if permitted)

CHAPTER 5 IMPLEMENTATION

5.1 Application Setup

The Product Registration Form, reports, dashboards, and workflows were configured to manage product product registrations and warranty claims.

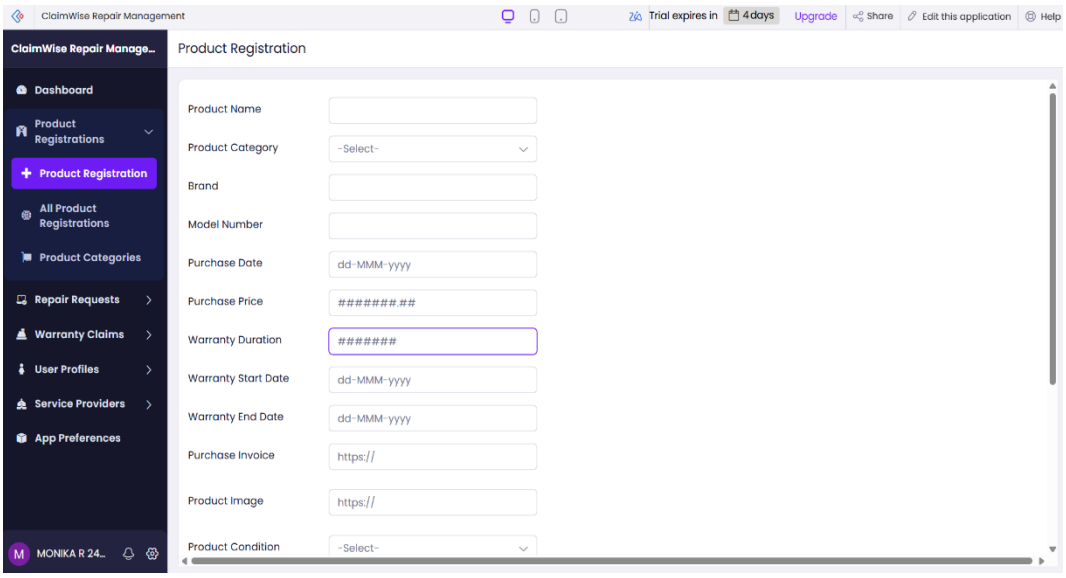


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

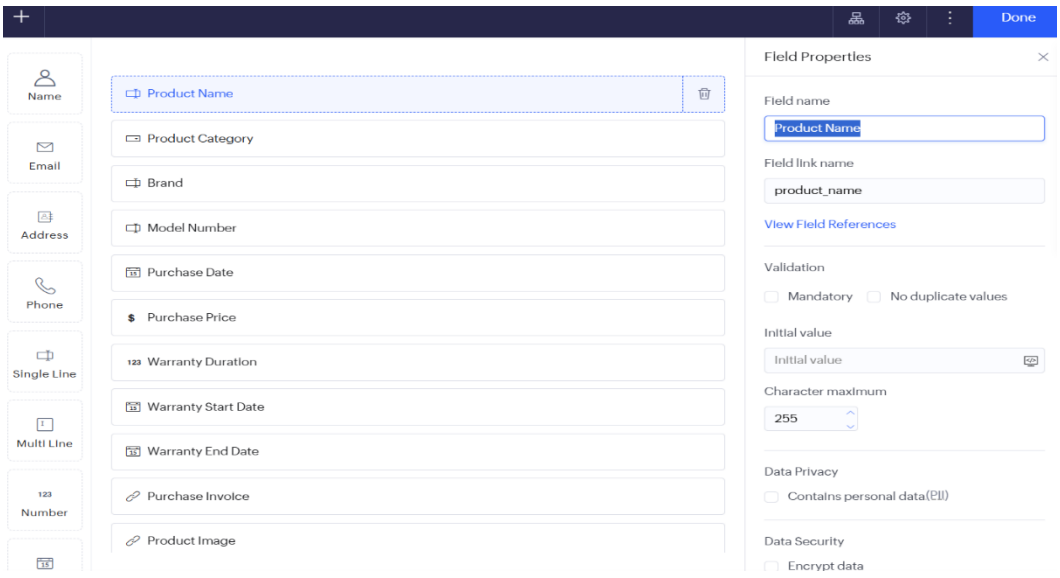


Figure 5.2: Form in the Form Builder

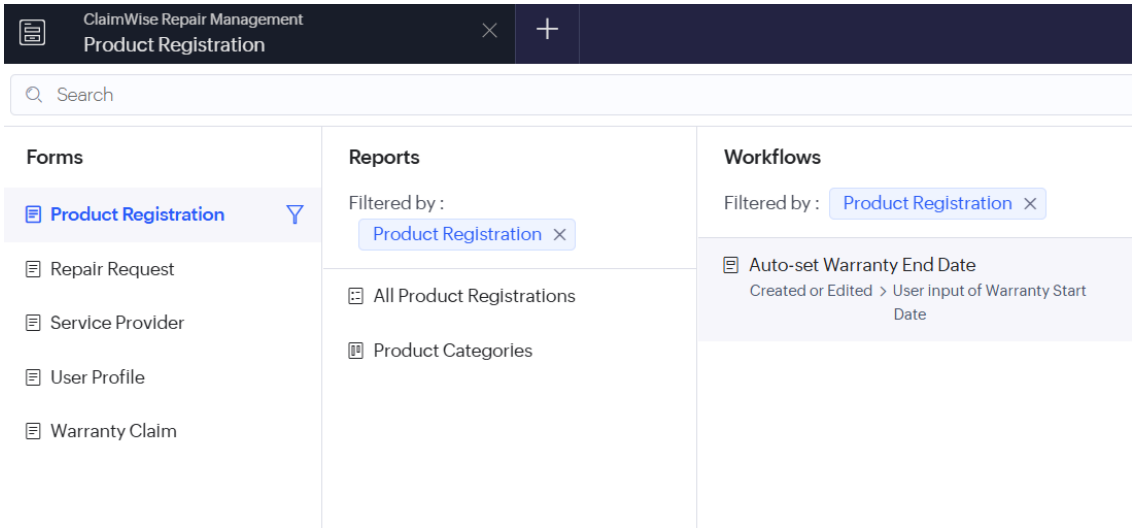


Figure 5.3: Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Product Registration Report	List	Product Registration	Displays registered product and warranty details	-	Administrator
Warranty Claim Report	List / Status view	Warranty Claim	Grouped or filtered by claim_status	-	Administrator
Repair Request Report	List	Repair Request	Displays repair request records	-	Administrator
Assigned Repair Report	List	Assigned Repair	Displays assigned service and repair details	-	Administrator

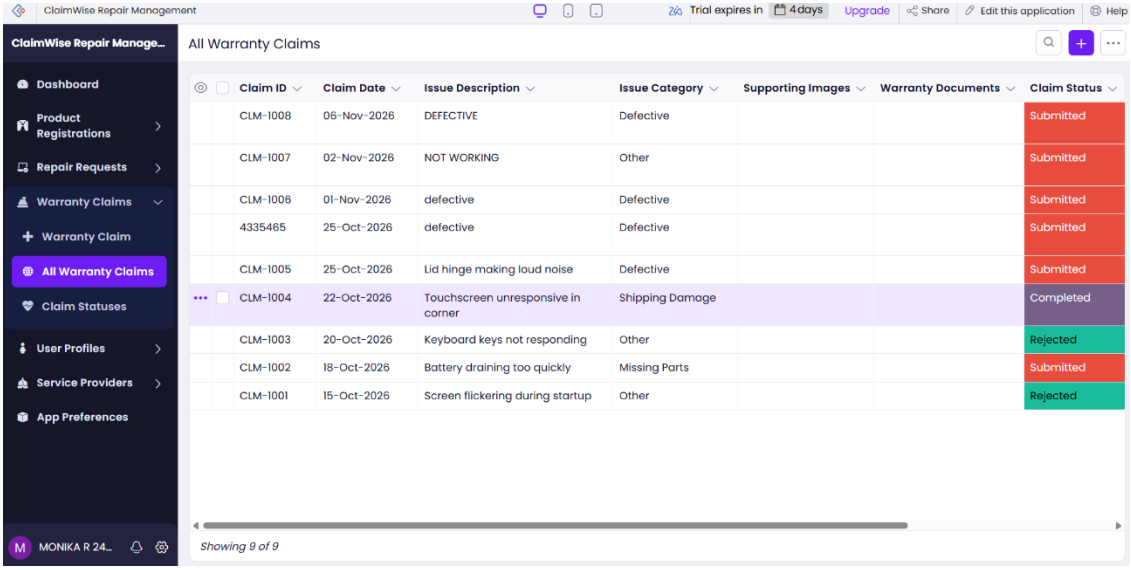


Figure 5.4: Warranty Claim Report

5.4 Pages and Dashboards

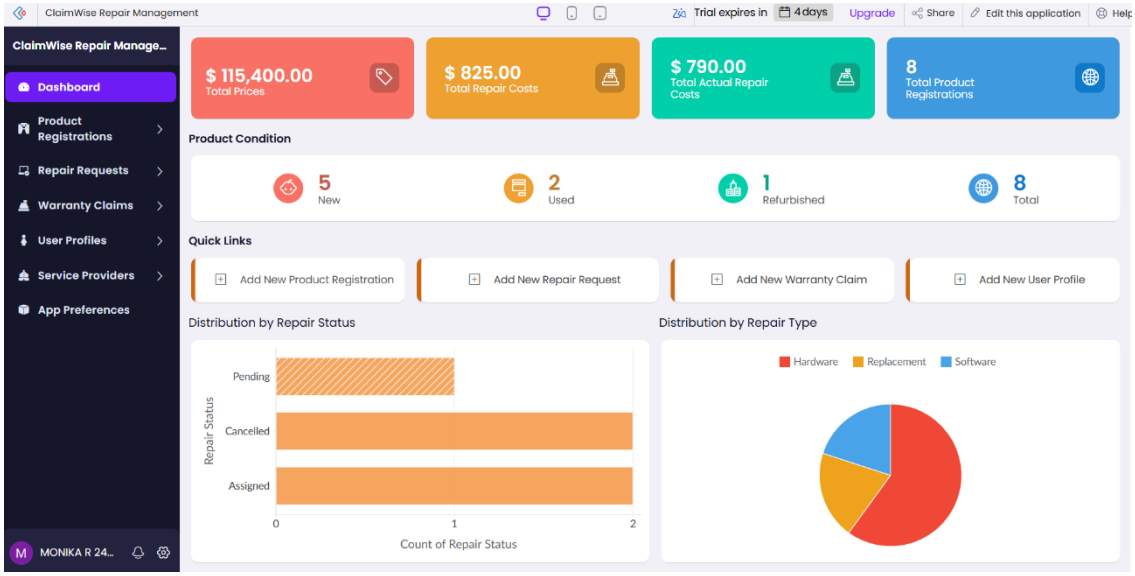


Figure 5.5: Dashboard Page Name

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Product Registration form event	Calculate / populate warranty end date where configured	Date calculation / field assignment
DS-02	Warranty Claim form event	Generate unique Claim ID	Record lookup / string handling
DS-03	Repair Request form event	Generate unique Repair ID	Record lookup / string handling

Code 5.1: Warranty / Repair Calculation Script

```
if(input.claim_status == null || input.claim_status == "")
{
        alert "Please select a claim status.";
        cancel submit;
}
```

Code 5.2: Claim Notification Workflow

```
warrantyClaimRecord = warranty_claim[ID == input.ID];
if(warrantyClaimRecord.claim_status == "Approved")
{
        if(linkedProductRecords.count() > 0)
        {
                registeredUser = linkedProductRecords.registered_user;
                if(!registeredUser.isNull())
                {
                        userProfileRecords = user_profile[ID == registeredUser];
                        if(userProfileRecords.count() > 0)
                        {
                                customerEmail = userProfileRecords.email;
                                sendmail
                                [
                                        from :zoho.loginuserid
```

```
                                        to :customerEmail
                                        subject :"Your Warranty Claim Has Been
Approved"
message :"Dear Customer,
            Your warranty claim has been approved. We will process your claim shortly.
Thank you for choosing our service.

Best regards,
“Warranty Support Team"
                            }
                    }
            }
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Warranty Reminder	Scheduled / form workflow as configured	Configured schedule or record event	Warranty date condition	Send reminder if enabled and tested	Verify final configuration
Admin New-Claim Notification	Form Workflow	New Warranty Claim created	New claim submitted	Notify administrator if enabled	Verify final configuration
Customer Claim-Status Notification	Form Workflow	Claim status updated	Approved / Rejected	Notify registered customer if enabled	Verify final configuration

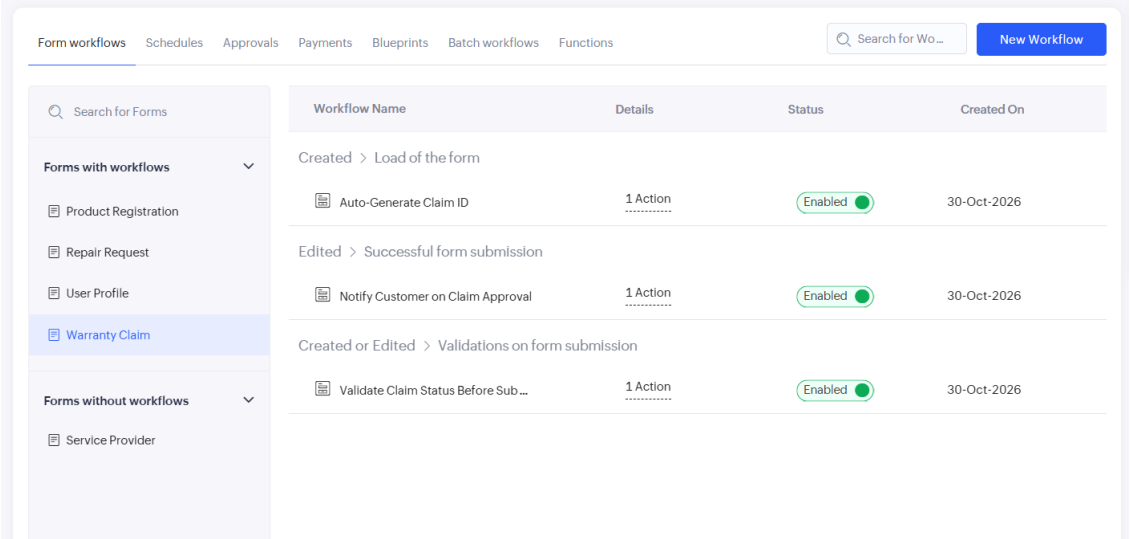


Figure 5.6: Warranty Reminder Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho CRM	Zoho CRM - ClaimWise Custom Tab	Import App from Zoho Creator	Zoho Account	Forms, reports, workflows and records are managed within Creator.
Email Notification	Zoho Creator Email Service	Workflow Email Action	Platform managed	Admin / customer notifications only after successful configuration and test.

Code 5.3: No external integration configured

```
// Integration: Idea2Impact with Zoho CRM
// Method: Import App from Zoho Creator
// Purpose: Make the Creator application accessible
// through the Idea2Impact custom tab in Zoho CRM.
// No custom Deluge invokeurl script used.
```

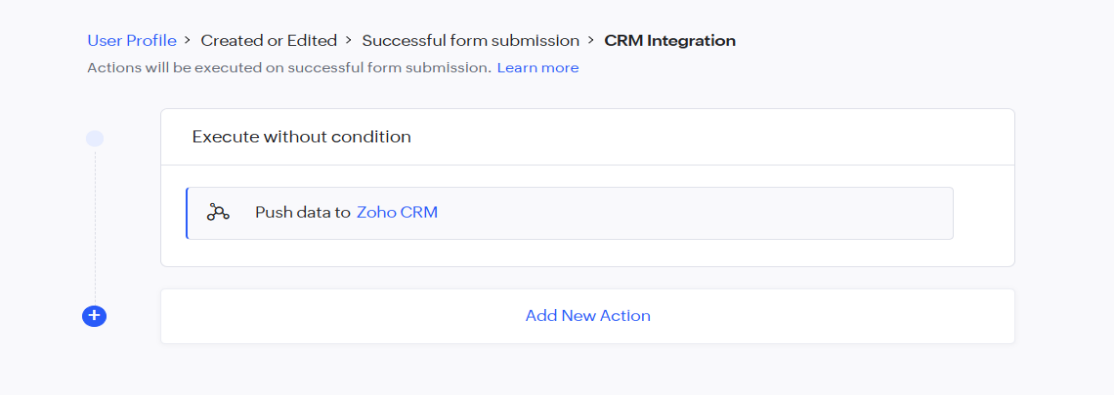


Figure 5.7: Connection Setup and Postman / Response Output

5.8 User Management and Role-Based Access

Users access ClaimWise through their registered Zoho accounts, with the administrator managing application records and workflows.
Role-based permissions are intended to restrict access to application data and administrative functions.

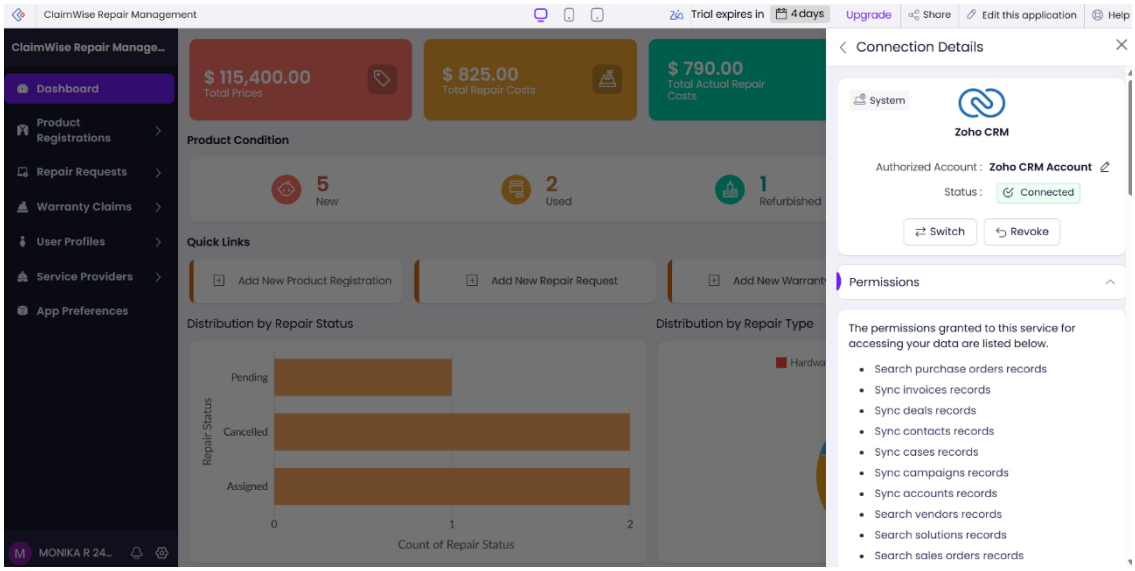


Figure 5.8: [Roles and Permission Profiles Configuration]

5.10 Pro-Code Extension

The ClaimWise application was tested to check the functionality of product registration forms, reports, and workflows. Form testing focused on entering and submitting customer details. Workflow testing checked initial status updates and email notifications. Reports and dashboards were checked to ensure submitted records were displayed correctly. The application features were reviewed from the administrator's perspective to verify proper operation.

5.10 Application on Mobile

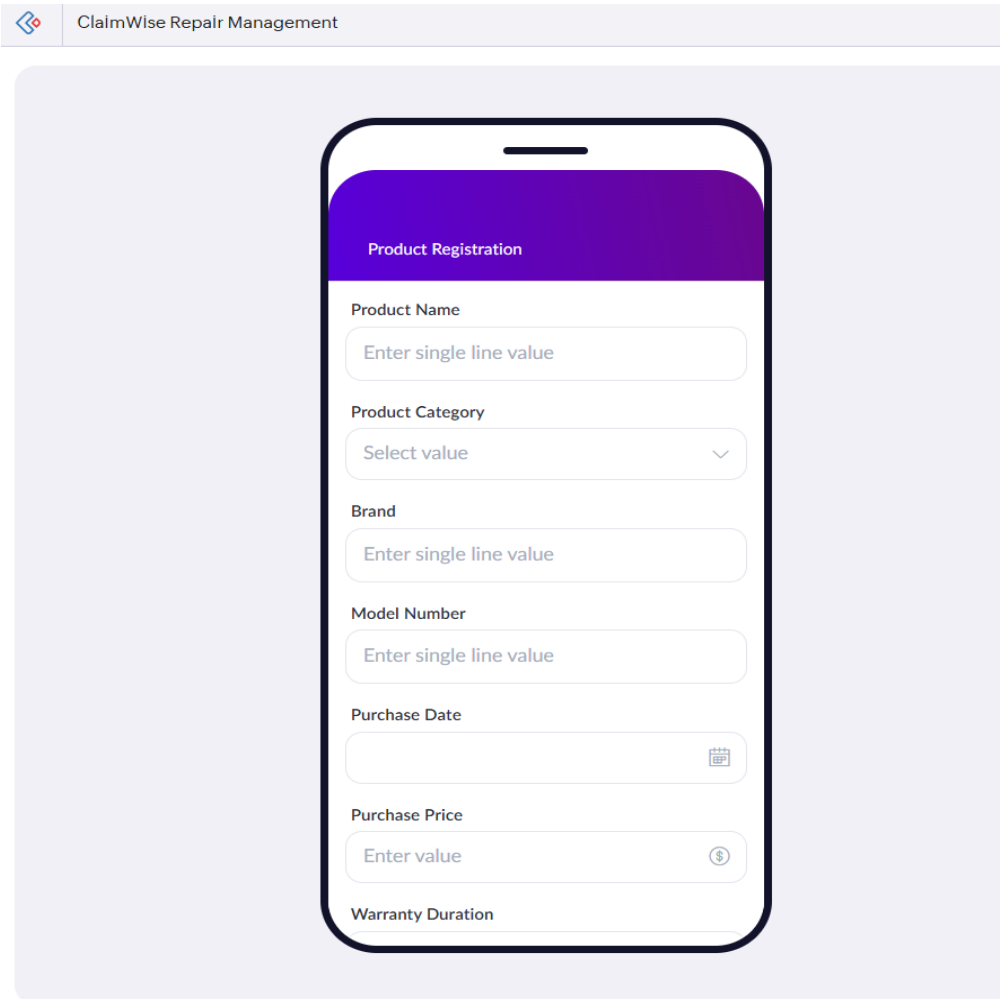


Figure 5.10: ClaimWise Application on Mobile

CHAPTER 6 TESTING AND DEBUGGING

6.1 Testing Strategy

The ClaimWise application was tested to check the functionality of product registration forms, reports, and workflows. Form testing focused on entering and submitting customer details. Workflow testing checked initial status updates and email notifications. Reports and dashboards were checked to ensure submitted records were displayed correctly. The application features were reviewed from the administrator's perspective to verify proper operation.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a product	Enter product name, warranty details and registered user; submit.	Product record is saved.	To be entered from actual test	Pass
TC-02	FR-02	Verify warranty end date	Enter the required warranty inputs and save.	Warranty end date is calculated / Stored as configured.	To be entered from actual test	Pass
TC-03	FR-03	Generate Claim ID	Submit a warranty claim with linked product.	Unique Claim ID is generated.	To be entered from actual test	Pass
TC-04	FR-06	Calculate repair cost difference	Enter repair cost values and save.	Cost difference is calculated.	To be entered from actual test	Pass

6.3 Debugging Deluge Scripts

Debugging was performed by checking workflow configurations, execution results, and error messages during form submission. The Deluge info statement helps display field values and trace script execution. Workflow logs can be checked to identify errors in automated actions.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Repair ID text value caused numeric conversion error.	Identifier field is Single Line text.	Used text-based identifier generation for the existing field.	Resolved
BUG-02	A workflow did not execute as expected during setup.	Trigger / field link name configuration.	Reviewed and corrected the workflow configuration.	Resolved

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Registered User	Product registration and claim submission can be demonstrated.	Review required fields and form usability.
User 2	Administrator	Dashboard helps monitor product, claim and repair records.	Verify displayed counts against reports.

CHAPTER 7 DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

ClaimWise was developed and configured using the Zoho Creator cloud platform. The application is accessed through a web browser using a registered Zoho account. The product registration form, reports, dashboards, and workflows were configured and tested in the Creator environment. The application can be accessed by authorized users through the web interface.

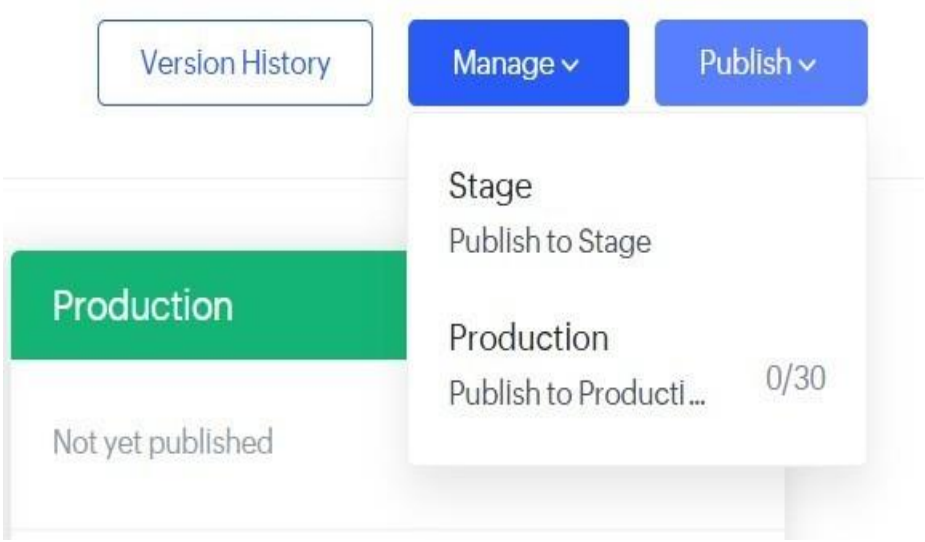


Figure 7.1: [Environments / Publish to Production]

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (☐)
1	All test cases executed in Development / Stage	☐
2	Sample/test data removed or isolated	☐
3	Roles, permissions and users configured in Production	☐
4	Connections authorised in Production	☐
5	Schedules and workflows enabled	☐
6	Application published and URL verified	☐
7	Version / release notes recorded	☐

7.2 Performance Optimisation

Centralized record management ClaimWise uses Zoho Creator reports and workflows to manage product and claim records efficiently. Status Report, spreadsheet, and calendar reports help organize and access application details. Reports and dashboards provide a centralized view of submissions.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Centralized record management	Registration Form	Manual records	Centralized records	Easier tracking
Status-based organization	Status-based organization	Manual tracking	Status-wise view	Faster monitoring
Dashboard reporting	Admin Dashboard	Manual analysis	Manual analysis	Easier analysis

7.3 Security Features

ClaimWise uses Zoho Creator's account-based access to provide controlled application access. Customer details and uploaded documents are maintained within the application. Administrative access is intended for authorized users to manage product registrations and claim statuses.

Table 7.3: **Security Controls Implemented**

Security Control	Where applied	Threat Mitigated
Zoho account authentication	Application login	Unauthorized login
User access permissions	Application access	Unauthorized data access
Mandatory field validation	Registration Form	Incomplete or invalid submissions
Platform-managed data security	Zoho Creator	Data exposure risks

CHAPTER 8 RESULTS AND DISCUSSION

8.1 Application Output

The ClaimWise application was tested using a sample product registration and warranty claim scenario. A customer submits an innovative product through the product registration form. The submitted details are stored and displayed in the product and claim reports. The application status is initially set to Submitted, and the administrator can monitor and update the claim status using the status-based report. The dashboard displays product, claim and repair statistics for easy monitoring.



Figure 8.1: Scenario Step 1

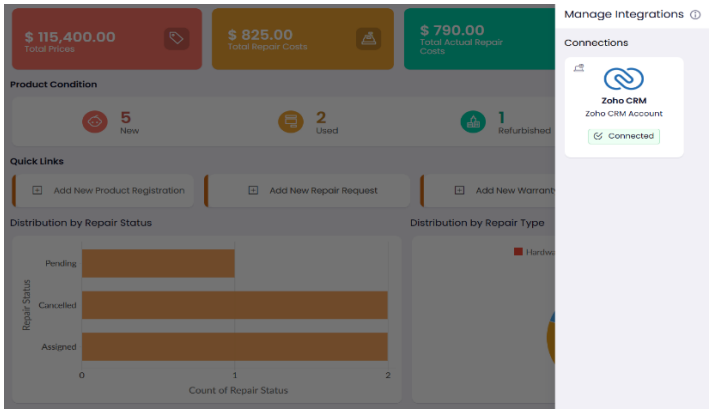


Figure 8.2: Scenario Step 2

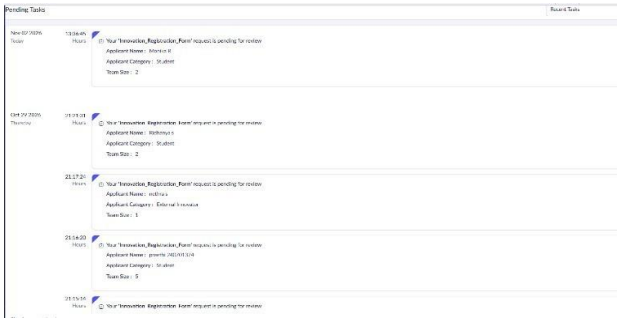


Figure 8.3: Scenario Step 3

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Create a centralized product registration and user profile system.	Section 5.2 – Forms	User Profile and Product Registration forms.
Manage warranty claims and generate unique identifiers.	Sections 5.2 and 5.5	Warranty Claim form and Claim ID generation.
Manage repairs and monitor key records.	Sections 5.2–5.4	Repair Request, Assigned Repair and dashboard indicators.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Warranty tracking	Manual checking of purchase and warranty details.	Product records with warranty end dates.	Easier warranty lookup.
Claim tracking	Claims maintained through separate communication.	Warranty Claim records with status.	Clearer claim monitoring.
Repair management	Repair details maintained separately.	Repair Request and Assigned Repair forms.	Centralized repair information.
Report generation	Manual record preparation.	Zoho Creator reports and dashboard counts.	Easier monitoring.

8.4 Limitations of the Low-Code Approach

Zoho Creator provides easy application development, but it has some limitations. Advanced UI customisation is restricted compared to traditional programming. Complex business logic may require additional Deluge scripting. API integrations may also be subject to platform limits.

CHAPTER 9 CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The ClaimWise application was successfully developed using Zoho Creator to simplify warranty claim submission and management. It provides product registration forms, reports, status-based reports, calendar reports, and an administrator dashboard to monitor applications. Workflows and email notifications help reduce manual effort and improve application tracking.

The project demonstrates how low-code development can be used to build an efficient warranty and repair management system. Zoho Creator application also provides access to the application within the Creator environment.

9.2 Future Enhancements

Integrate the future AI-based issue analysis for AI-based analysis and evaluation of submitted products and claims.

Develop a dedicated mobile application and customer portal for easier access.

Add advanced analytics to identify recurring product issues, monitor claim resolution time and compare repair costs.

9.3 Learning Outcomes and Career Relevance

This project provided practical experience in developing applications using Zoho Creator, designing forms, creating reports, and configuring workflows. I gained knowledge of Deluge scripting, application integration, data management, and dashboard creation. These skills are useful for career paths such as low-code developer, citizen developer, solutions consultant, and automation engineer. The project also improved my problem-solving and application development skills.

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[1] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/. [Accessed: 3-Nov-2026].

[2] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/. [Accessed: 3-Nov-2026].

[3] N. Prinz, C. Rentrop, and M. Huber, “Low-Code Development Platforms – A Literature Review,” in *Proceedings of the Americas Conference on Information Systems (AMCIS)*, 2021.

[4] K. Rokis and M. Kirikova, “Exploring Low-Code Development: A Comprehensive Literature Review,” *Complex Systems Informatics and Modeling Quarterly*, no. 36, pp. 68–86, 2023, doi: 10.7250/csimq.2023-36.04.

[5] M. O. Ajimati, N. Carroll, and M. Maher, “Adoption of Low-Code and No-Code Development: A Systematic Literature Review and Future Research Agenda,” *Journal of Systems and Software*, vol. 222, Art. no. 112300, 2025, doi: 10.1016/j.jss.2024.112300.

APPENDIX A ZOHO CREATOR FEATURE COVERAGE CHECKLIST.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections/menu organised	Y	ClaimWise – Repair & Warranty Assistant	5.1	Fig. 5.1
II	Forms with multiple field types	Y	User Profile, Product Registration, Warranty Claim, Repair Request, Service Provider, Assigned Repair	5.2	Fig. 5.2
II	Lookup field (relationship between forms)	Y	registered user and linked_product lookup fields	4.2, 5.2	Fig. 5.2
II	Subform (parent–child data)	N	Not implemented / not confirmed	4.2, 5.2	-
II	Field validations / layout rules	Y*	Configured fields and form settings; verify final rules	5.2	Fig. 5.2
II	List reports and status-based views	Y	Product, warranty claim and repair reports	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	N*	Only include views actually configured in the application	5.3	-
II	Dashboard charts / indicators	Y	ClaimWise dashboard record-count indicators	5.4	Fig. 5.5
II	Page / dashboard built with UI Builder	Y*	Dashboard page; verify exact builder configuration	5.4	Fig. 5.5
II	Deluge – form events and calculations (validate / success / field actions)	Y	Warranty end date, Claim ID, Repair ID and repair cost calculations script	5.5	Code / workflow w screenshot
II	Deluge – functions, fetch, loops, collections	N	Not confirmed as implemented	5.5	-
II	Email / in-app notifications	To verify	Admin and customer notification workflows; verify successful delivery Workflow	5.6	Workflow screenshot

II	Form workflow	Y*	Configured form workflows; verify each is enabled and tested	5.6	-
II	Scheduled workflow	To verify	Warranty reminder schedule; verify trigger and delivery	5.6	-
II	Approval process	N	Manual administrator claim status update	5.6	-
II	Blueprint	N	Not implemented / not confirmed	5.6	-
II	Batch / report workflow	N	Not implemented / not confirmed	5.6	-

APPENDIX B COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture and business value of low-code application development using Zoho Creator.	Ch. 1, 2	High	
CO2	Develop business applications using Zoho Creator forms, reports, workflows and Deluge scripting.	Ch. 4, 5, 6 5.6), 6	Medium	
CO3	Use relationships, platform services, notifications and access controls in a Creator application. Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.2, 4.6, 5.6–5.8	Medium	
CO4	Deploy, secure and manage Zoho Creator applications following governance practices.	Ch. 7	Low	
CO5	Evaluate low-code development strategies and future application enhancements.	Ch. 8, 9 8.4, 8.5, 9.3	Medium	

APPENDIX C COMPLETE DELUGE SOURCE CODE

C.1 Set Claim Status Script ID: DS-01
Location: Product Registration / Warranty Claim form event Purpose: Set the claim status after a new application is submitted.

DS-01: Set Claim Status input.claim_status = "Submitted"; info "Initial claim status set to Submitted"

C.2 Display Customer Information Script ID: DS-02
Location: Product Registration / Warranty Claim form event Purpose: Display customer information in the execution log for debugging. DS-02: Customer Information Debugging info "Customer Name: " + input.customer_name; info "Email Address: " + input.email; info "Application submitted successfully";

C.3 Repair ID Generation Script ID: DS-03 Location: Product Registration form validation
Purpose: Generate a unique Repair ID for a repair request. Paste the exact saved and tested Deluge script here.

{
[Insert the final Repair ID generation script from Zoho Creator.]
}

APPENDIX D USER MANUAL

D.1 Accessing the Application

Open Google Chrome or another web browser.

Visit the Zoho Creator website: https://www.zoho.com/creator/.

Log in using your registered Zoho account.

Open the ClaimWise application.

For mobile access, use the Zoho Creator mobile app and log in with your Zoho account.

D.2 Instructions for Customer

- Open the Product Registration Form.
- Enter the product details and warranty information.
- Upload the required documents and product cover image.
- Submit the warranty claim and note the generated Claim ID.
- Check the application details and status through the available reports.

D.3 instructions for Administrator

• Log in and open the ClaimWise application.

Open the Product Registration Report to view submitted applications.

Use the Warranty Claim Status Report to monitor review stages.

Update the application status when required.

Open the Admin Dashboard to view product, claim and repair statistics and charts.

Use the spreadsheet and calendar reports to monitor applications.

APPENDIX E EVALUATION RUBRIC

Student Name	MONIKA R	Register No.	240701332
Project Title	CLAIMWISE – REPAIR & WARRANTY ASSISTANT	Date	10-11-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	MONIKA R	Register No.	240701332
Department / Section	CSE	Date of Viva	10-11-2026
Application URL	To be entered	Demo Duration	---------- min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

EXAMINAR'S REMARKS

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____